Validate

System architecture and design

How the system is built, how it reaches decisions, and how to use it. Explained in plain language.



Version 1.0 · November 2026

A receipt is a claim, not a confirmation.

WELCOME

Contents

This document explains how Validate is built and how it works. It avoids technical language. Where a technical word is unavoidable, it is explained where it first appears and again in the glossary on the last page.

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PART 1

Understand

Why the system exists, what it does and the promises it keeps

pages 3 to 8

PART 2

How it works

Follow a payment and a person through the checks, step by step

pages 9 to 16

PART 3

Use it

Hands-on guides with screenshots and a quick reference

pages 17 to 20

How to read it. If you only have ten minutes, read pages 4, 5 and 9 to understand what the system does and the promises it keeps, then go straight to the how-to guides on pages 17 to 19.

THE PROBLEM

Why Validate exists

Many organisations take payments: school fees, course fees, membership fees. Customers often prove they have paid by sending a photo or screenshot of a receipt. Someone then has to look at it and decide whether the money really arrived.

1 Receipts are easy to fake

A photo of a receipt can be edited in seconds. The same receipt can be reused for several bills. To the eye, a good fake looks real.

2 Checking by hand is slow

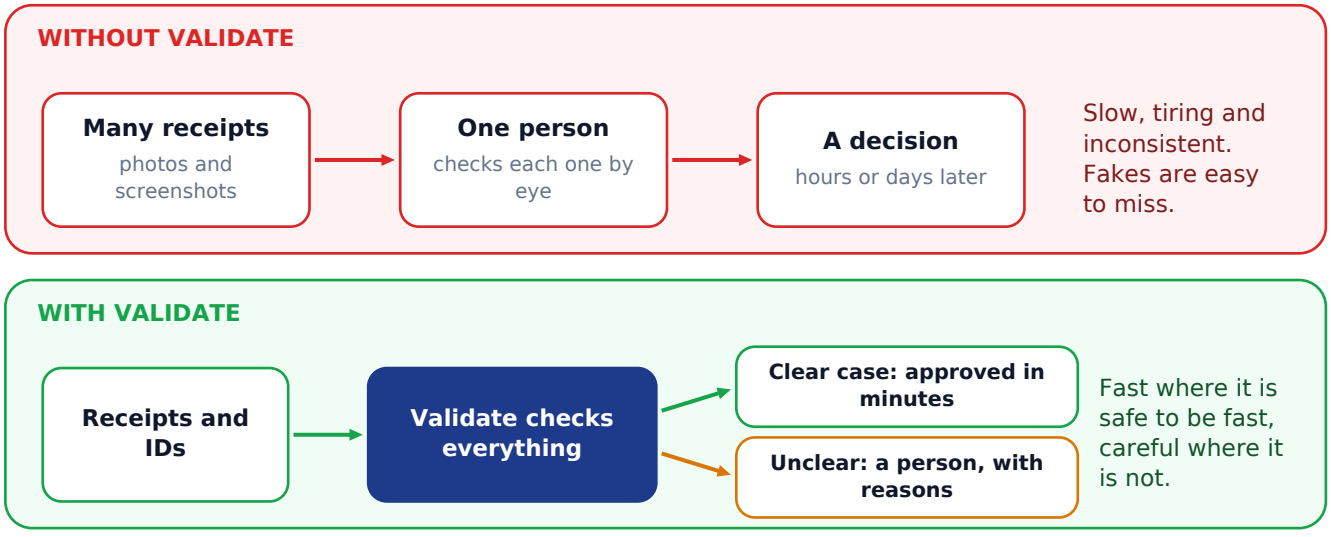
Queues build up. People who have paid wait for access. Staff spend their day on repetitive checks instead of on people who need help.

3 Knowing who someone is

Identity documents are checked by eye, differently by each person, and the photos are often kept in places they should not be.

What a good solution looks like

- **Quick** when the case is clear, so honest people are not kept waiting.
- **Careful** when the case is not clear, with a person making the call.
- **Fair and explainable.** Every decision has a reason you can read.
- **Safe with personal information.** Only what is needed, only for as long as needed.



In one sentence: Validate checks payments and identities, approves the clear cases on its own, and hands everything doubtful to a person together with the reasons.

AT A GLANCE

What Validate does

Validate does two jobs. Both follow the same pattern: someone sends in a photo, the system reads it and checks it, and the result is one of a small number of clear outcomes.

Job 1: Check a payment

The question: did the money really arrive, and does it match the bill?

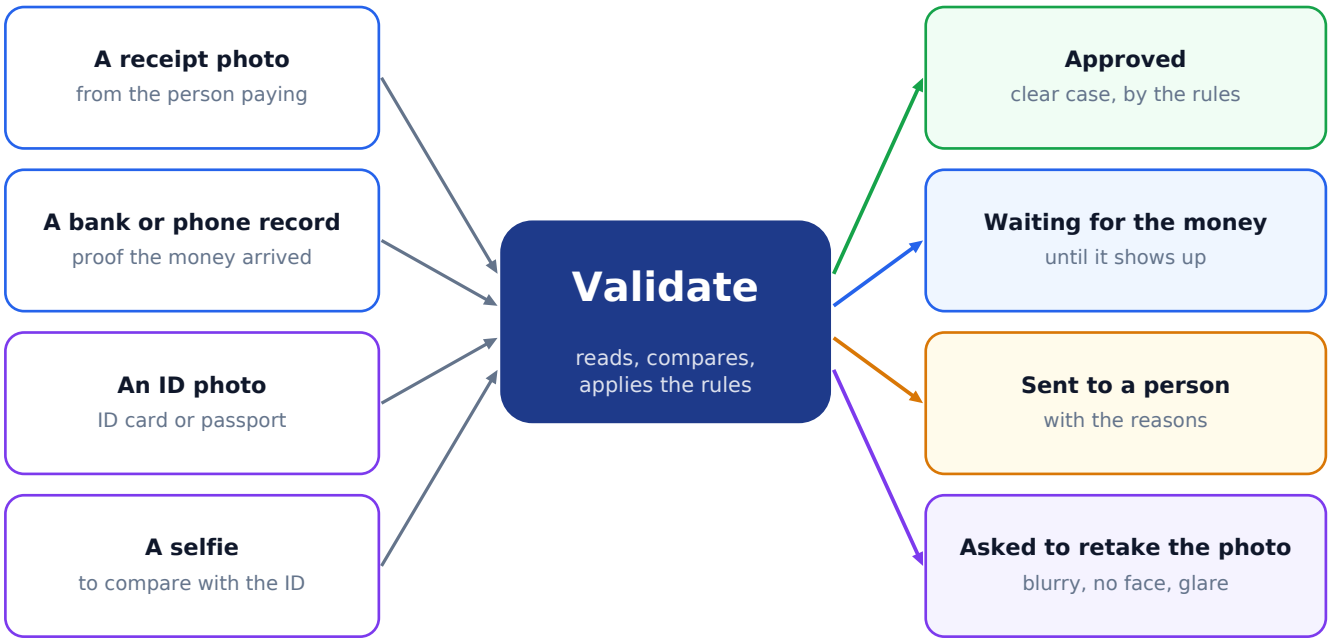
The person sends a photo or message showing they paid. The system reads it, then looks for the matching money in the bank records. A receipt alone is never enough.

Job 2: Check a person

The question: is this person who they say they are?

The person sends a photo of an ID card or passport and a selfie. The system reads the document, compares the face with the selfie, and runs a set of safety checks.

What goes in and what comes out



The three kinds of result

Approved by the rules

Every check passed. No person had to touch it. A sample is checked by a person afterwards.

Sent to a person

Something was unclear or looked wrong. A reviewer sees all the evidence and the reasons.

Asked to retake

The photo could not be used. The person is told, in plain words, how to fix it.

Important: the system never turns anyone away on its own. A refusal is always a person's decision.

PRINCIPLES

Our design promises

Every part of the system is built around a few promises. They explain why it behaves the way it does, and they are the best way to judge whether it is behaving properly.

1 A picture is only a claim

A receipt can be edited in seconds, so on its own it proves nothing. Only the money itself, seen in a bank record or an alert from the receiving phone, settles a payment.

2 The computer reads. Rules and people decide.

Software is used to read photos and compare faces, which it is good at. It is never allowed to decide on its own. Decisions come from written rules or from a person.

3 When unsure, a person decides

Any doubt, any missing information, any broken part, and the case goes to a person. The system is built to fail safely towards a human.

4 Nobody is rejected by a machine

The system can approve clear cases. It cannot reject. Turning someone down, or revoking an approval, always needs a named person and a written reason.

5 Everything is written down

Every decision is recorded: who or what decided, when, and why. The record cannot be changed afterwards. Corrections are added as new lines, never by editing old ones.

6 Off until you switch it on

Automatic approval and automatic identity checks are switched off at the start. An organisation turns them on deliberately, and one master switch can turn them off everywhere at once.

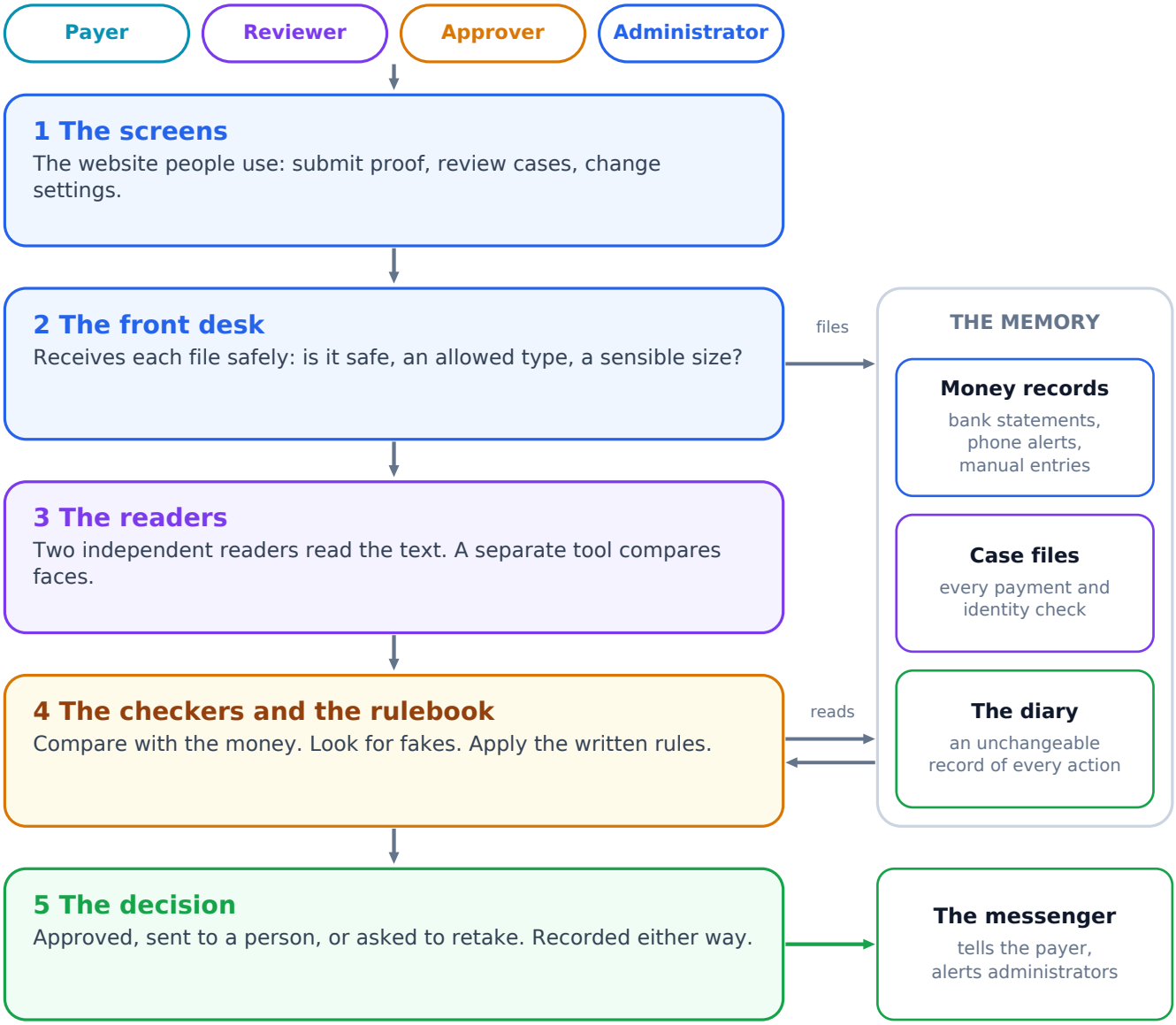
Why these promises matter. Payment fraud and identity fraud both work by making something look more trustworthy than it is. The system is designed so that looking trustworthy is never enough: it needs independent proof, and a person is always the final safety net.

“A receipt is a claim, not a confirmation.”

ARCHITECTURE

The big picture

Think of a busy, well-run office. There is a front desk, two clerks who read documents separately, a rulebook, a manager, a filing room, and someone who tells people the outcome. Validate is built the same way, in layers.



How information flows

A file enters at the top, is checked and read as it moves down, and comes out as a decision at the bottom. Each layer only does its own job and hands over to the next.

Why layers matter

If one layer is unsure or fails, the case drops back to a person. No single part, and especially not the readers, can approve something by itself.

ARCHITECTURE

The parts of the system

Nine parts work together. Each one has a plain-language comparison so you can picture what it does.

1 The screens

The website people use. Each person sees only what their role needs.
Like the windows and counters of a service office.

2 The front desk

Receives photos and documents. Scans them for harmful files, refuses unusable ones, limits how many can be sent.
Like a receptionist who checks what is handed over.

3 The readers

Two independent readers pull the facts out of a picture: amounts, dates, names, numbers. A face tool compares a portrait with a selfie.
Like two clerks who read the same page separately.

4 The money records

Bank statements, alerts from the phone that received the money, and manual entries. This is the proof that money really arrived.
Like the cashier's own record book.

5 The rulebook

A written list of conditions. Automatic approval only happens when every line is satisfied. Organisations can adjust the dials.
Like a policy manual that is applied the same way every time.

6 The fake spotter

Looks for warning signs: edited amounts, reused receipts, editing marks, pictures that do not match the money.
Like a careful auditor.

7 The review desk

The waiting list for people. Each case shows the pictures, what was read, the scores and the reasons it needs a person.
Like a manager's in-tray with the file attached.

8 The diary

An unchangeable record of every decision and action, with who, when and why.
Like a signed logbook where pages cannot be torn out.

9 The messenger

Tells payers where their case stands, in neutral words, and warns administrators when something needs attention.
Like a helpful notice board and a phone call.

ARCHITECTURE

Who does what

There are four kinds of user. The system enforces what each one may do, so the permissions do not depend on anyone remembering them.

Each role includes everything the role before it can do.

Payer

- Sends proof of payment
- Sends ID and selfie
- Sees the result of their own case, in plain words

Reviewer

- Everything a payer can do
- Works the waiting list
- Confirms a matching payment
- Asks for a clearer photo
- Verifies or sends back an identity

Approver

- Everything a reviewer can do
- Gives the second approval on large amounts
- Reverses a mistaken approval
- Records money received outside a statement
- Reviews staff identity checks

Administrator

- Everything an approver can do
- Sets the rules and limits
- Sees suspected fakes and decides
- Sees how much is at risk
- Switches automation on or off

Built-in safeguards between people

Nobody reviews their own work

A person cannot approve a case they submitted, or review their own identity check.

Large amounts need two people

Above a set amount, two different approvers must both agree before money is allocated.

Payers are not accused

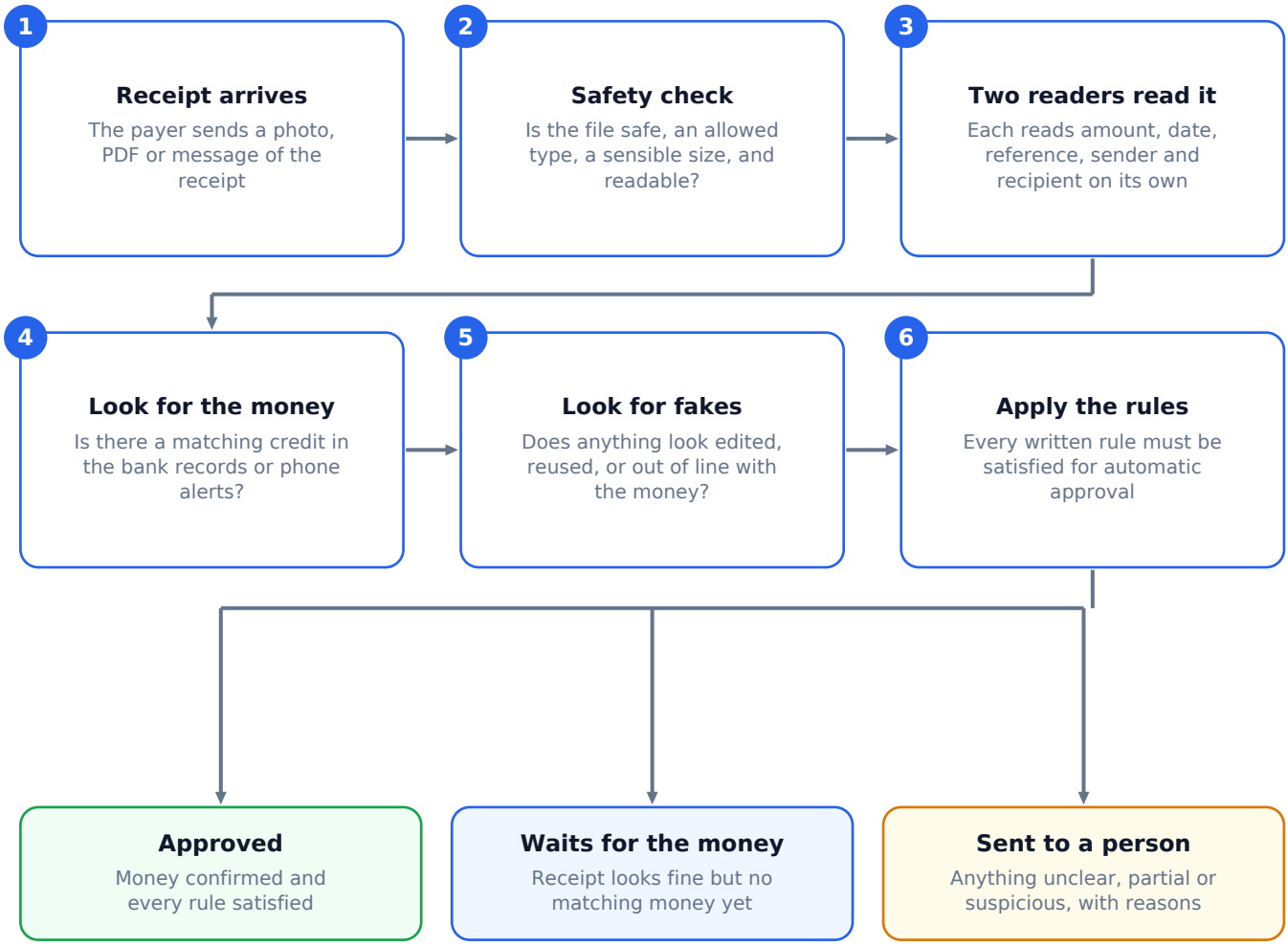
If a case looks suspicious, the payer is only told it needs a manual check. The reasons are for staff.

Strong sign-in for staff. Reviewers, approvers and administrators are expected to sign in with two-step verification when the system is connected to an organisation's own sign-in.

HOW IT WORKS

How a payment is checked

Follow one payment from start to finish. Suppose Amara pays a USD 40 course fee by mobile money and sends in her receipt.



What makes this safe

Steps 4 and 5 look past the picture. The receipt says what Amara claims; the bank record says what actually happened. Approval needs the second to agree with the first.

If the money is not there yet

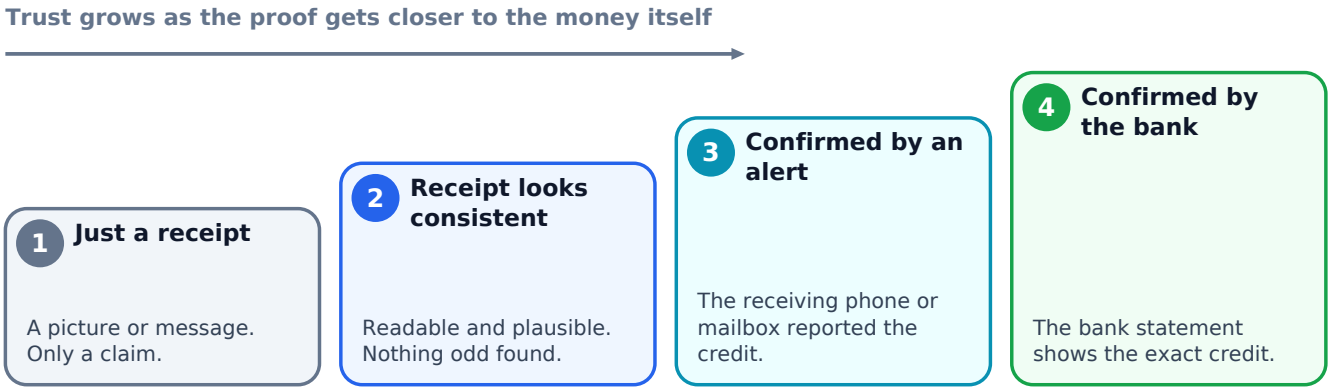
The receipt waits. When the bank statement arrives, the waiting receipt is matched automatically and the rules run again. Nobody has to resubmit.

What Amara is told: only the outcome, in simple words, such as “Your payment is confirmed” or “Your payment is waiting for confirmation from our bank records”. She never sees internal scores or suspicions.

HOW IT WORKS

Telling real from fake

Because a receipt can be faked, the system asks a different question: where could the proof of the money come from? There are three trustworthy sources, and the more reliable the source, the more the system is willing to do on its own.



Three trusted sources of proof

- **Bank statement.** The most reliable. The credit appears in the organisation's own account.
- **Alert from the receiving phone or mailbox.** A message forwarded the moment the money lands.
- **A person's manual record.** For money that arrived another way. Large amounts need a second person to confirm.

Warning signs the system looks for

Warning sign	What it means in plain words	Weight
Amount differs from the bank	The receipt says one amount, the money received is another. A classic sign of editing.	High
Same receipt used twice	The same file or reference is attached to a second bill.	High
Looks computer-generated	Hidden marks suggest it was made by an image generator rather than a phone.	High
Editing software left marks	The file's hidden details show it passed through a photo editor.	Medium
A patch pasted into the image	Part of the picture has a different noise pattern from the rest.	Medium
Date in the future, or wrong recipient	The payment date cannot be right, or the money went to a different account.	Medium
Payer history	This payer has had suspicious submissions confirmed before.	Medium

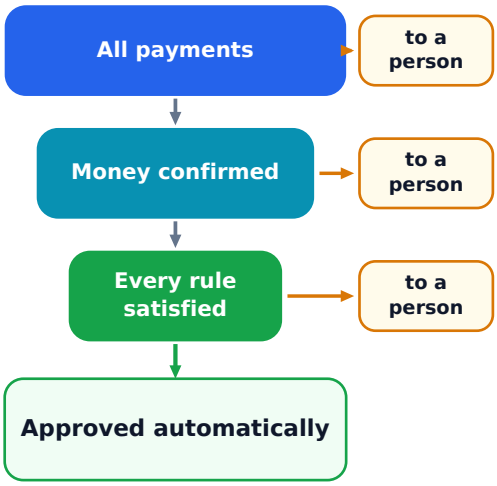
What happens if it looks fake. The case is held, an administrator is alerted with the reasons in plain language, and the payer is told only that the payment “needs a manual check”. A person decides. A warning sign alone never rejects anyone.

HOW IT WORKS

Approving the clear cases automatically

Most payments are ordinary. Automatic approval exists so that ordinary cases are settled in minutes, and people's time is kept for the unusual ones. It is a fixed checklist, not a judgement call.

	Every one of these must be true
✓	Automatic approval is switched on for this organisation, and not switched off globally
✓	The payment method is one the organisation has chosen to automate
✓	The money is confirmed by a bank statement or a phone alert, not only by the receipt
✓	Nothing looks suspicious, and no fraud alert is open
✓	The amount is under the organisation's limit for automatic approval
✓	The payment settles the bill exactly: not part-paid, not overpaid
✓	No person has already started looking at it
✓	The hard checks are run again at the moment of approval
✓	If required by the organisation: the payer's identity is verified



Keeping watch on the automation

1 A sample is checked by a person

A set share of automatic approvals (for example one in five) is put in front of a reviewer, who marks each as correct or overturned.

2 The overturn rate is the dial

If reviewers rarely disagree, the limits can be raised carefully. If they often disagree, the limits come down. The decision is based on evidence.

3 Every approval can be undone

An approval is an ordinary record, not a one-way door. An approver can reverse it, with a written reason that goes in the diary.

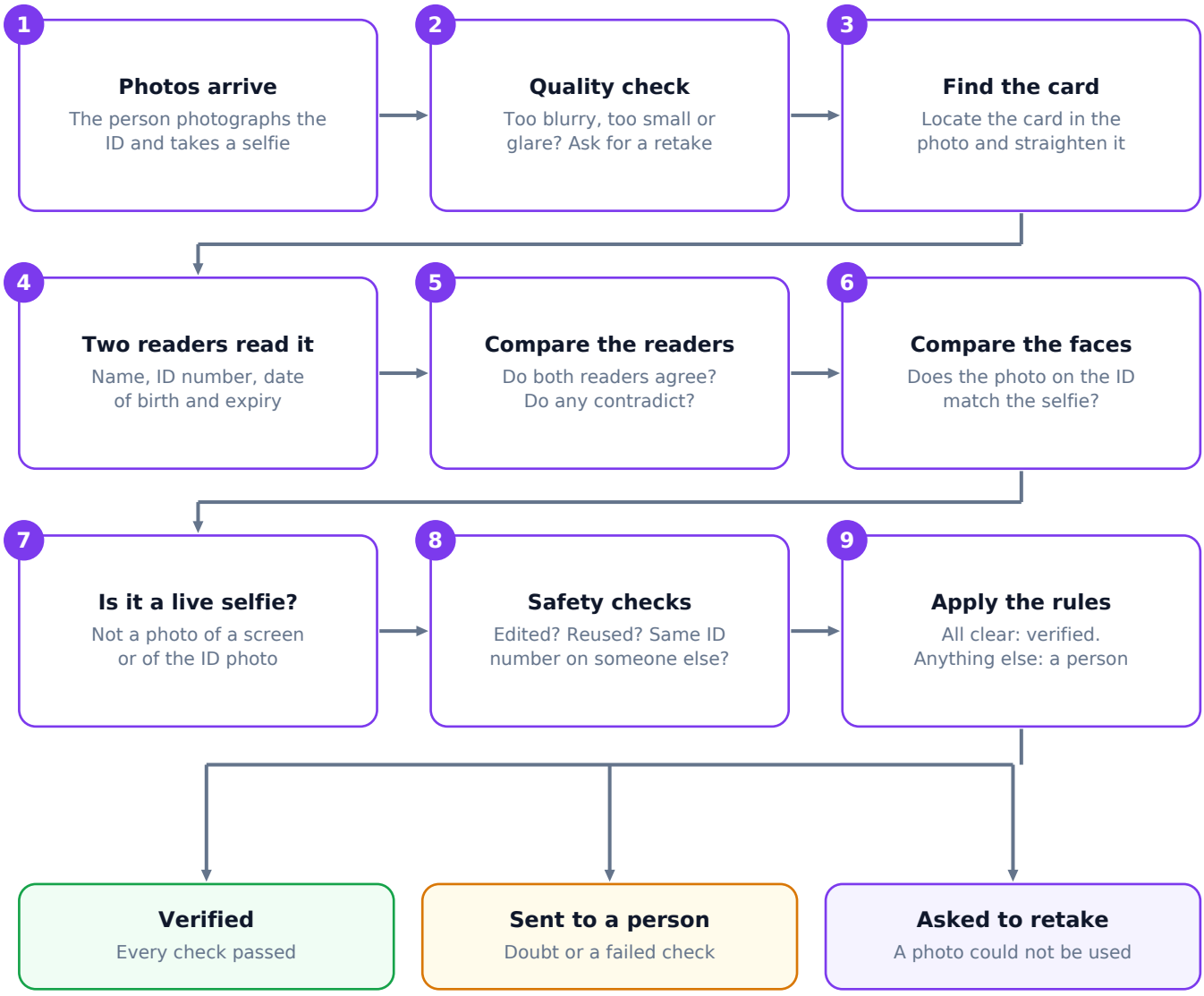
Safe by design. The checklist is short enough to read in a minute and is applied identically every time. The software that reads receipts has no say in the decision: it only supplies facts to the checklist.

Switching it on gently. An organisation can first run the checklist in watching mode, where the system records what it would have decided without acting. Weeks later the results are compared with what people decided, and only then is it switched on.

HOW IT WORKS

How a person is checked

Checking an identity follows the same shape as checking a payment, with different questions. Here is the journey from the photos to the result.



Two readers, one answer

Two independent readers read the document. A field is trusted only when both agree. A passport also has a coded strip with built-in self-checking numbers, which catches edited or misread details.

Never an automatic refusal

If a face does not match, a document has expired or two readers disagree, the case goes to a person. The system never turns someone away on its own.

HOW IT WORKS

What the identity checks look for

Ten questions are asked of every identity check. Any one that fails sends the case to a person with that reason on the page.

Check	What it asks	Why it matters
Photo quality	Is the photo sharp, big enough, free of glare?	A poor photo cannot be read safely. The person is asked to retake it.
Document type and country	Is it an accepted kind of document from an accepted country?	The organisation chooses which documents it can verify with confidence.
Readers agree	Do the two readers agree on name, number, birth date, expiry?	Disagreement can mean a misread, or an edited document.
Coded strip (passports)	Do the check numbers add up?	They are very hard to fake. An edited date breaks the sums.
Not expired	Is the document still valid, with some time left?	An expired document cannot confirm identity.
Face match	Does the photo on the ID look like the selfie?	Stops someone using another person's ID.
Live selfie	Does the selfie look like a real person in front of a camera?	Stops photos of screens or printed pictures.
Not reused	Have these photos, or this ID number, been used by someone else?	One ID should belong to one person.
No signs of editing	Does the image show signs of being altered?	Catches pasted or changed details.

Privacy by design

Photos are stored privately and deleted after a set time (90 days unless the organisation chooses otherwise).

The ID number is kept only as a scrambled fingerprint, used to spot the same ID used twice.

Face measurements are never stored.

Who sees what

Only reviewers can see the photos, and every viewing is written in the diary.

The person who submitted sees only the status of their own check.

A verification expires after a year and the person is asked again.

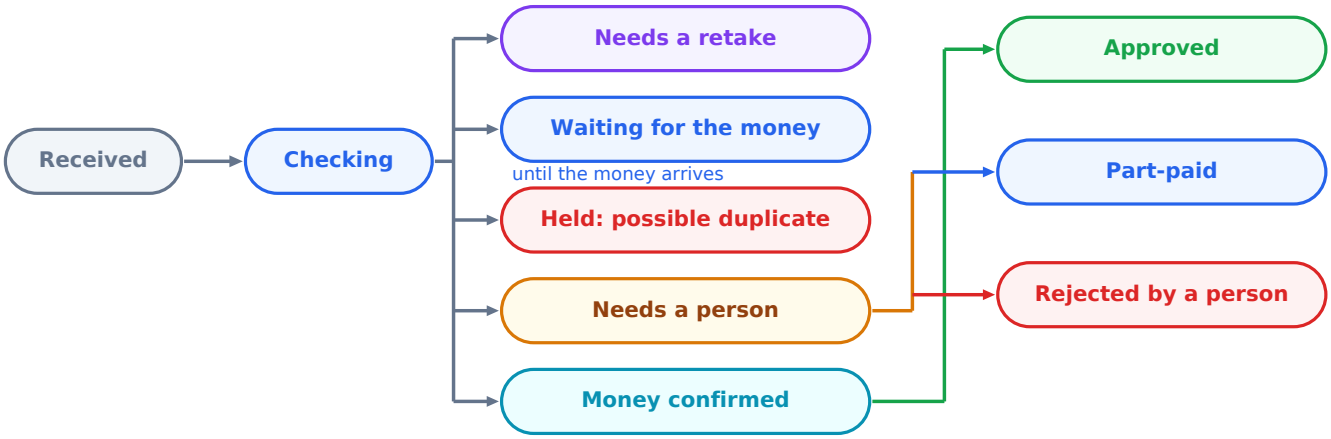
The same promise applies. The software reads and compares. Fixed rules or a named person decide. When anything is unclear, a person gets the full picture, not a bare “failed”.

HOW IT WORKS

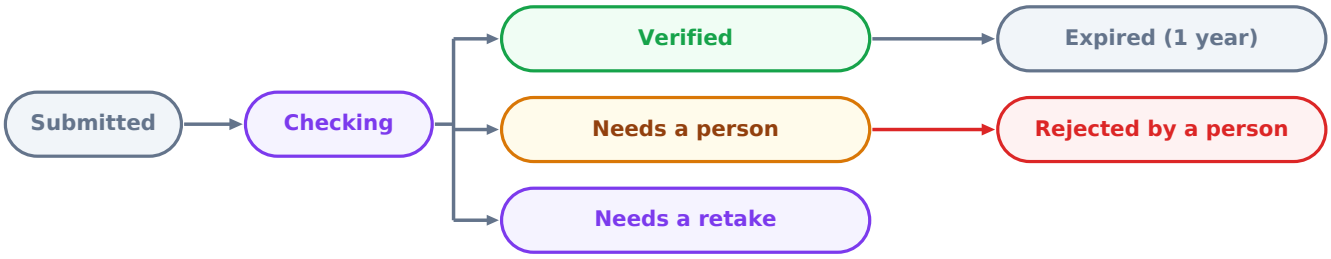
When the system is not sure

Every payment and every identity check is always in one clear state. These are the states and the ways a case moves between them. Almost everything that is not a clear approval ends up with a person.

A PAYMENT



AN IDENTITY CHECK



What each state means for the person who sent the proof

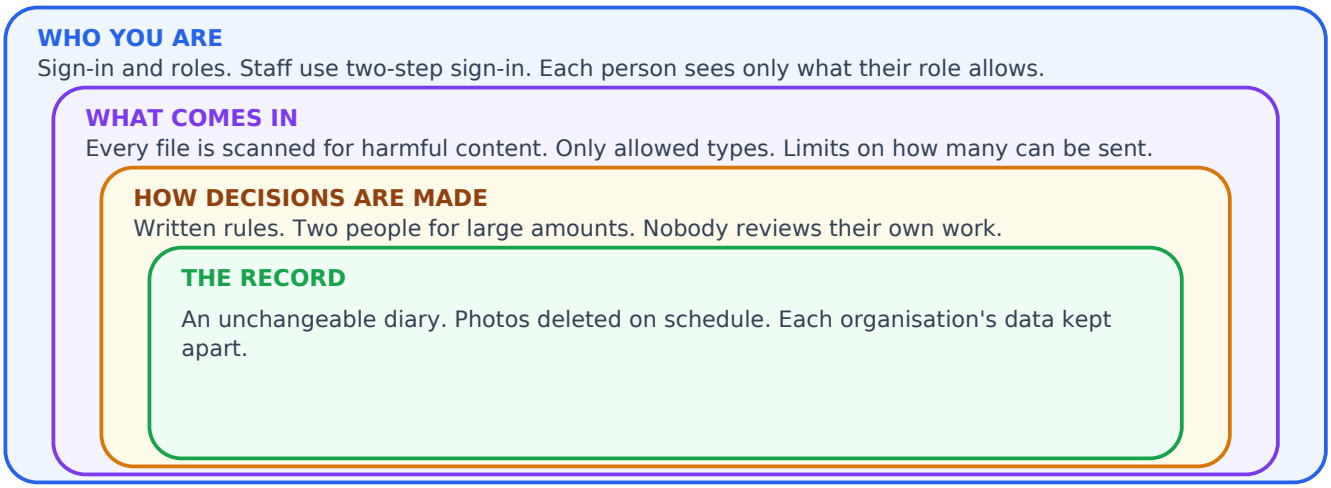
State	What the person is told	Who acts next
Waiting for the money	Your payment is waiting for confirmation from our bank records.	The system, when the money arrives
Needs a person	Your documents are with a reviewer, who will confirm shortly.	A reviewer
Needs a retake	The photo is blurry or too small. Please retake it in good light.	The person
Held: possible duplicate	Your payment needs a manual check by our finance team.	An administrator

Held does not mean accused. The words shown to the person are deliberately neutral. The reasons, scores and pictures are shown only to staff.

HOW IT WORKS

Records and keeping information safe

A system that handles money and identity documents has to be trustworthy in how it keeps records and protects information. Four layers of protection sit around every case.



1 A diary that cannot be rewritten

Every decision is saved with who or what made it, when, why, and which version of the rules applied.

The record is locked: nothing can be changed or deleted afterwards. A correction is a new entry that refers to the old one, so the full history is always visible.

2 Protecting the people behind the pictures

Photos are viewed through short-lived links, and each viewing is recorded.

Personal details are not written into logs.

Photos are deleted after the retention period. Only the fact that they existed remains.

3 Guarding against abuse

Limits stop a flood of uploads from one person.

Reused photos, reused IDs and edited documents are detected.

Large amounts and sensitive actions need extra people.

4 If a part breaks

A broken or missing reader never approves anything.

After a few failed attempts a case is handed to a person.

A forgotten case is picked up again automatically.

A note on rules and law. How long personal information may be kept, and what people must be told, depends on local law. The retention time is a setting so each organisation can set it to what applies to them. This document is not legal advice.

HOW IT WORKS

Settings, switches and honest limits

The organisation, not the software, decides how much to automate. Three modes make it safe to introduce gradually, and a single master switch can stop all automation at once.

OFF

People decide everything. The checks still run, so reviewers see the scores and reasons.

DEFAULT

WATCHING

The system records what it would have decided but changes nothing. Used to measure before trusting it.

RECOMMENDED FIRST STEP

ON

Clear cases are approved by the rules. A sample is checked by a person. Nothing is ever rejected.

AFTER TESTING

Setting (in everyday words)	What it controls	Starting value
Automatic approval	Whether clear payments are approved without a person, and for which payment methods	Off
Amount limit	The most that may be approved automatically, per currency	Small amount
How sure the face match must be	How closely an ID photo must resemble the selfie	Must be tuned
Which countries and documents	Where ID documents can be verified automatically	None until chosen
How long photos are kept	Days before identity photos are deleted	90 days
How long a verification lasts	Days before a person is asked to verify again	1 year
Require identity before approval	Whether a payer must be verified before payments auto-approve	Off
Share checked by a person	How many automatic decisions are sampled for review	5 to 25 percent

Honest limits: what to know before relying on it

Face matching needs testing

How well faces are matched depends on the tool used. It must be tested on your own kinds of ID photos before automatic checks are switched on.

The live-person check is basic

It catches simple tricks such as a photo of a screen. It is not a defence against a determined, sophisticated attack.

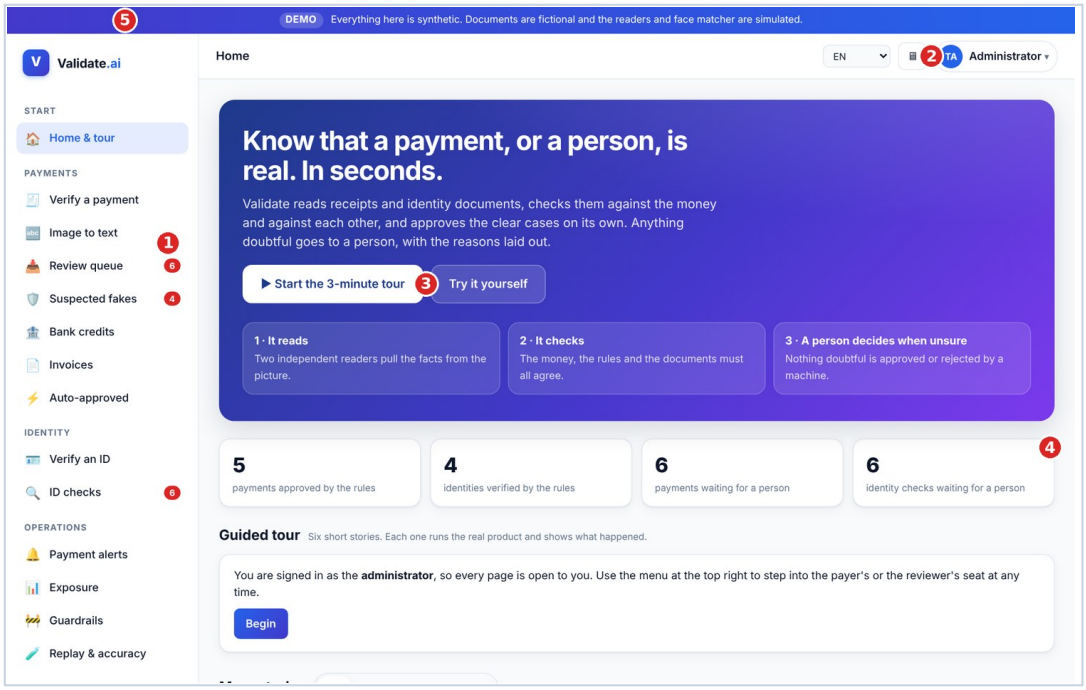
The demonstration is simulated

In the demonstration version the readers and face matching are stand-ins that only understand the sample documents shown.

The system supports decisions; it does not replace judgement. Where a case needs understanding of someone's circumstances, a person must make that call.

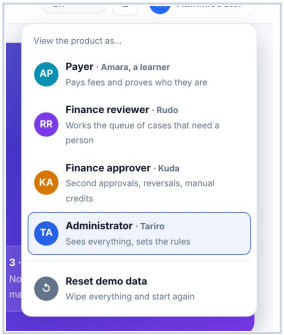
HANDS-ON GUIDE

How to use it: getting started



The home screen. The numbers match the list below.

#	What you see	What it is for
1	The menu on the left	Moves between pages. You only see the pages your role allows.
2	Your name and role, top right	Click to step into another seat: payer, reviewer, approver or administrator.
3	“Start the 3-minute tour”	Walks through six short stories, each run with a click.
4	Live numbers	How many cases were approved automatically and how many wait for a person.
5	The blue banner	A reminder that the demonstration uses made-up people and documents.



Switching seats

Click your name at the top right to see the four seats. Pick one and the pages change to match what that person can do. In the demonstration this is the quickest way to see the system from every point of view.

The same menu has **Reset demo data**, which clears everything and starts again with fresh sample cases.

Start

Home page and tour

Payments

Verify a payment, review queue,
suspected fakes, bank credits,
invoices, auto-approved

Identity

Verify an ID,
ID checks (review desk)

Operations

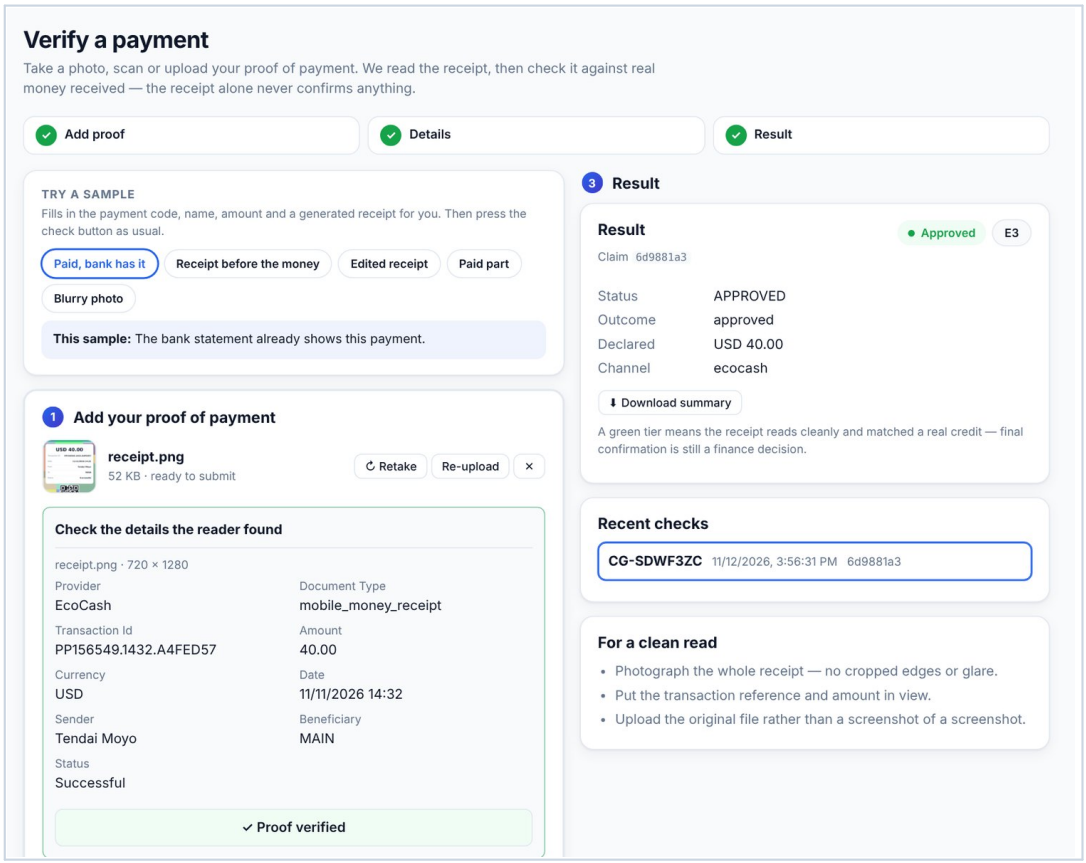
Payment alerts, exposure,
guardrails, replay

HANDS-ON GUIDE

How to use it: checking a payment

This guide follows one payment from the payer's seat to the reviewer's desk. In the demonstration there is a “Try a sample” panel, so you do not need a real receipt.

1. **Choose your seat** — Top right: pick “Payer”
2. **Open the page** — Menu: Verify a payment
3. **Add the proof** — Try a sample, or upload a receipt
4. **Press the button** — “Validate this payment”
5. **Read the result** — Approved, waiting, or a person



Verify a payment: the sample is loaded on the left, and the result (Approved) appears on the right.

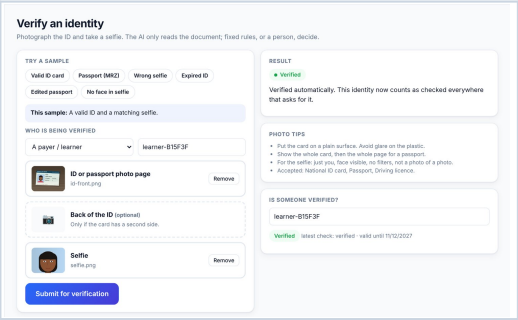
Result shown	What it means	What you do
Approved	The money was found and every rule passed.	Nothing. You are done.
Waiting for the money	The receipt looks right but no matching money is on record yet.	Wait. It is matched automatically when the money arrives.
Needs a person	A reviewer has to look, for example a part-payment.	Nothing. A reviewer takes it from here.
Needs a retake	The photo is too blurry or small.	Take a clearer photo and send it again.

As the reviewer: switch to “Finance reviewer”, open Review queue and click a case. The page shows the receipt, what was read, the matching bank record and a suggested next step. To confirm a payment you type the last four characters of the bank reference, which proves you checked the source.

HANDS-ON GUIDE · INCLUDING THE REVIEW DESK

How to use it: checking a person

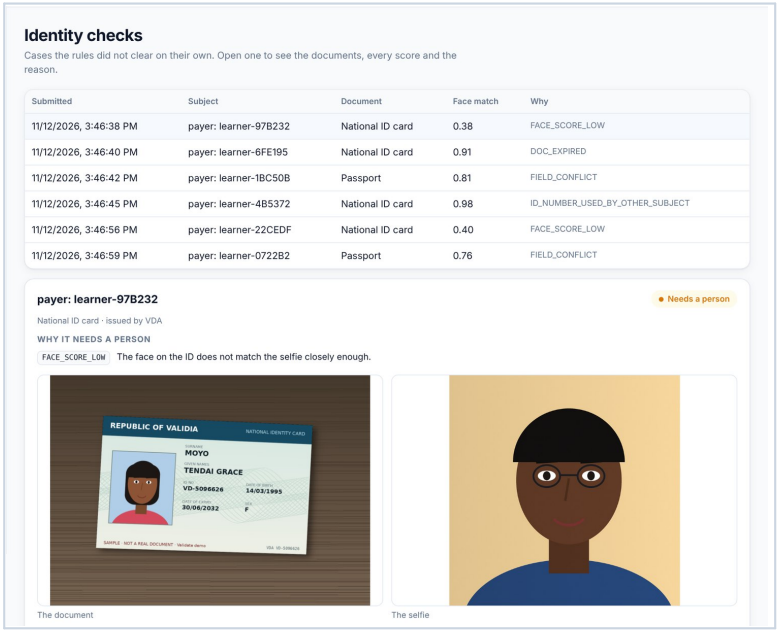
1. **Choose your seat** — Top right: pick "Payer"
2. **Open the page** — Menu: Verify an ID
3. **Add the photos** — Try a sample, or add ID and selfie
4. **Submit** — "Submit for verification"
5. **Read the result** — Verified, or with a reviewer



The payer's view: a sample is loaded and the result is Verified.

For the person sending the photos

They add the ID and a selfie and press submit. Within seconds they are told whether they are verified, or that a reviewer will look at it.



The reviewer's view of a case that needs a person.

On the review desk	What to do
Read **Why it needs a person** at the top	It gives the reason in plain words, for example that the face does not match closely enough.
Compare the document with the selfie	They are side by side. Trust your own eyes: the scores support you, they do not replace you.
Look at the scores and the list of checks	Green ticks passed. Anything amber or red is where to look first.
Decide	**Verify** if you are satisfied. **Ask for new photos** if a photo is unusable. **Reject** only with a written reason, which goes in the permanent record.

Good to know. Photos are only visible to reviewers. Every time you open one, the system writes it in the diary. A verification lasts a year, then the person is asked again.

HANDS-ON GUIDE

Quick reference and glossary

Who can see which page

Page	Payer	Reviewer	Approver	Admin
Home and tour	✓	✓	✓	✓
Verify a payment	✓	✓	✓	✓
Verify an ID	✓	✓	✓	✓
Review queue, suspected fakes, bank credits		✓	✓	✓
ID checks (review desk)		✓	✓	✓
Second approvals, reversals			✓	✓
Exposure, guardrails, replay, settings				✓

Stories in the guided tour

Story	What happens
Pay and get approved	Money already in the bank: approved in seconds
Receipt before the money	Waits, then is approved once the money arrives
Edited receipt	Receipt says 400, bank says 40: held and flagged
Verify an ID in seconds	Matching ID and selfie: verified automatically
Someone else's selfie	Face does not match: sent to a reviewer
Edited passport	Printed date disagrees with the coded strip: sent to a reviewer

Glossary and quick fixes

Word	Plain meaning
Claim	A statement that something is true, such as a receipt. Not yet proof.
Evidence	Something that supports a claim: a photo, a bank record, an alert.
Reviewer	A person who works the waiting list and makes decisions the system cannot.
Rules	The written checklist that decides what may be approved automatically.
Verified	An identity check that passed, either by the rules or by a reviewer.
Retake	A request to send a clearer photo. It is not a refusal.
The diary	The unchangeable record of every decision and action.
Watching mode	Automation records what it would decide, but does not act.

If this happens	Try this
A page is empty	Run a story from Home, or use “Reset demo data” in the top-right menu.
A menu item is missing	Your seat may not allow it. Switch seat from the top-right menu.
A sample says “expired”	Samples have a short life. Click the sample button again.
Everything looks odd	Reset the demonstration data from the top-right menu.

Starting the demonstration. Whoever sets it up runs a single command (./scripts/demo.sh), then opens the address it prints in a web browser. Nothing else needs installing, and all the people and documents shown are made up.